



Image Inpainting on the Edge

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Create an app that allows users to
remove objects from images and
fill gaps using generative models.

Problem Statement

Motivation

- ▶ To learn and apply the concepts of edge computing.
- ▶ Use the Qualcomm innovators development kit as a springboard to dive into the world of edge computing.
- ▶ Using state of the art hardware provided by Qualcomm to run AI models, especially image inpainting models, at the edge, using the CPU, GPU and the NPU (Hexagon Tensor Processor) or DSP (Digital Signal Processor).
- ▶ Get a taste of the exhilarating and enthralling world of embedded systems.

Methodology: Models

AOT-GAN

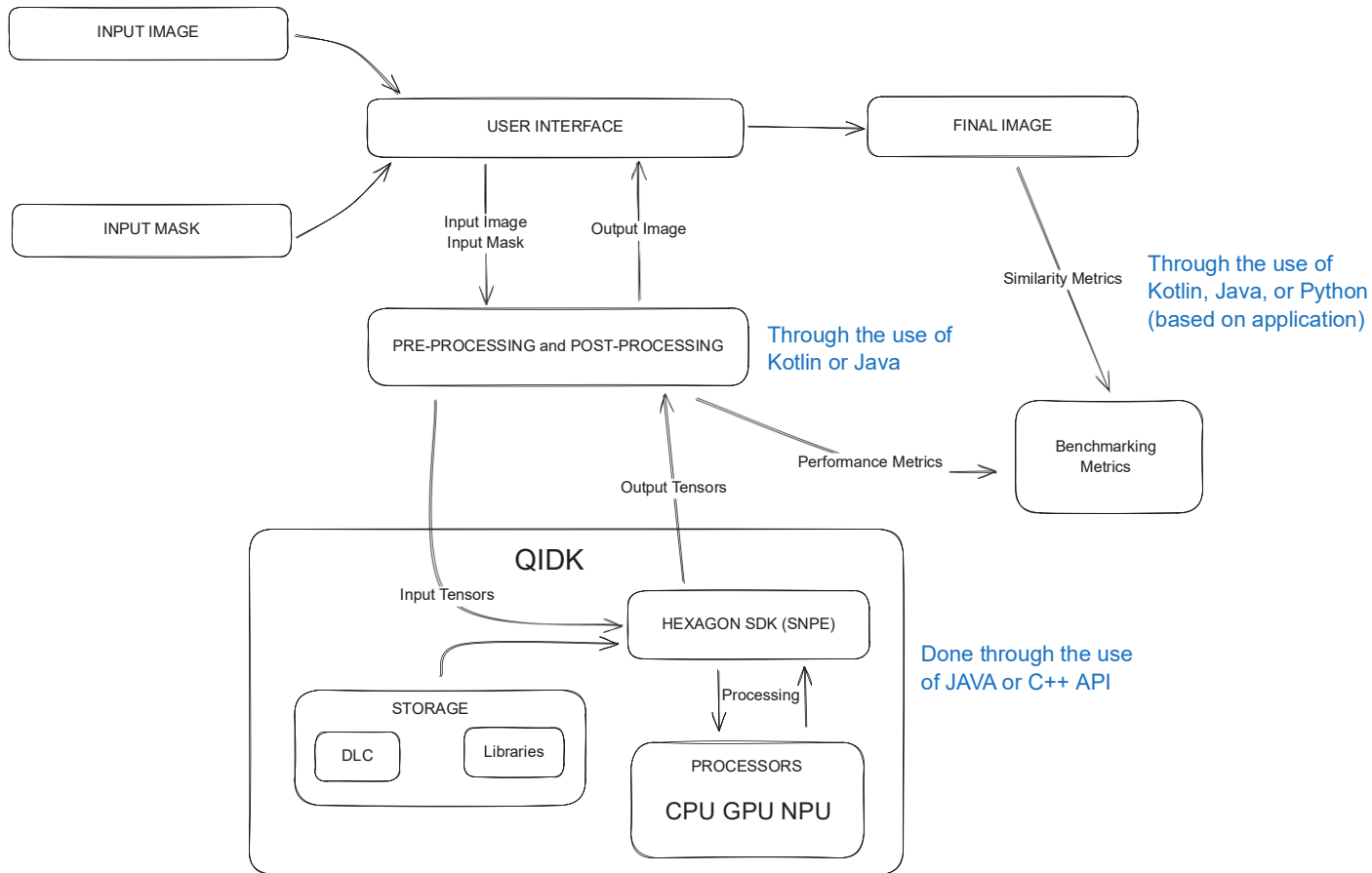
- ▶ Available on Qualcomm AI Hub
- ▶ Mentioned Metrics:
 - ▶ 120ms inference time
 - ▶ 8.35 inferences/s throughput
 - ▶ 0-157MB memory usage
 - ▶ 245 NPU Layers
 - ▶ Unquantized
 - ▶ Dimensions:
 - ▶ Image: (1x512x512x3)
 - ▶ Mask: (1x512x512x1)
- ▶ <https://aihub.qualcomm.com/models/aotgan>

LaMa-Dilated

- ▶ Available on Qualcomm AI Hub
- ▶ Mentioned Metrics:
 - ▶ 48.5ms inference time
 - ▶ 20.6 inferences/s throughput
 - ▶ 1-122MB memory usage
 - ▶ 340 NPU Layers
 - ▶ Unquantized
 - ▶ Quantized by us
 - ▶ Dimensions:
 - ▶ Image: (1x512x512x3)
 - ▶ Mask: (1x512x512x1)
- ▶ https://aihub.qualcomm.com/models/lama_dilated?searchTerm=ama

Framework

- ▶ Qualcomm SDK
 - ▶ Snapdragon Neural Processing Engine (SNPE)
 - ▶ TF-Lite Delegate
 - ▶ snpe-net-run
 - ▶ snpe-dlc-quant/snpe-dlc-quantize
- ▶ Android Framework
 - ▶ Kotlin/Java for routing and logic
 - ▶ C++/Java for the SNPE API
 - ▶ Gradle for management of dependencies
 - ▶ XML and Jetpack compose for UI
- ▶ Python (For similarity metrics)



Block Diagram

Parameters Considered



Performance Metrics

Inference Time

Memory Usage

CPU/GPU/NPU usage



Accuracy Metrics

Mean Squared Error (MSE)

Peak Signal to Noise Error (PSNR)

Structural Similarity Index Measure (SSIM)

Learned Perceptual Image Patch
Similarity (LPIPS)

Fréchet inception distance (FID)

Datasets

- ▶ Pictures taken by us (for testing)
- ▶ Image masks obtained from defacto-inpainting
 - ▶ <https://www.kaggle.com/datasets/defactodataset/defactoinpainting>
- ▶ Image JPG obtained from MSCOCO 2017 dataset
 - ▶ <https://www.kaggle.com/datasets/awsaf49/coco-2017-dataset>
- ▶ Utilized the above datasets to create a quantized model
- ▶ RAWs: generated from the above by us using python

Data Pre-Processing

- ▶ Conversion of Image for (jpg or png) to a tensor or RAW of the form (1,512,512,3) depending on the usage
 - ▶ Converted to RAW using a python script for use in snpe-bench and snpe-net-run
 - ▶ Converted to Tensors by the following methods
 - ▶ In the JAVA API, utilized pre-built functions for help in conversion
 - ▶ In the C++ API, manually loop through the pixels to convert the image to a tensor (Binary form) of the form mentioned above using Kotlin.
- ▶ In the case of pre-defined mask, we also ensure the orientation of the mask and the image align

Data Post-Processing

- ▶ In the C++ API app created outputs a tensor that is similar to the input tensor we give, hence we just convert the given binary tensor by doing the reverse of what we did for the input.
- ▶ For the Java API, we utilized the same functions utilized for the input here to ensure proper conversion.
- ▶ For snpe-net-run, we created another file which would convert from the raw format to a readable jpg or png as the output.

Demo: Workflow (Pre NPU)

Workflow 1 (on app)

- ▶ Pre-process image and mask to the respective input tensors
- ▶ Run the DLC on the provided input tensors
- ▶ Post-process the output tensor into an output image
- ▶ Output the in-painted image, and performance metrics to the screen.
- ▶ Take the ideal image, and the inpainted image and pass it to python (on laptop) to get similarity scores

Workflow 2 (using SNPE)

- ▶ Pre-process image and mask to the respective raw files
- ▶ Run the DLC on the provided input tensors, using snpe-net-run and snpe-bench.py.
- ▶ Post-process the output raw into an output image
- ▶ Get the csv file from snpe-bench.py
- ▶ Take the ideal image, and the inpainted image and pass it to python to get similarity scores

Working on the NPU

- ▶ After multiple attempts (mentioned previously) we have found a pipeline that runs our models directly on the NPU from an app.
- ▶ Workflow:
 - ▶ Pre-process image and mask to the respective input tensors
 - ▶ Utilize the TF-Lite file of the model and run it on the NPU by using TF-Lite Delegate to delegate the TF-Lite model to run on the NPU
 - ▶ Post-process the output tensor into an output image
 - ▶ Output the in-painted image, and performance metrics to the screen.
 - ▶ Take the ideal image, and the in-painted image and pass it to python (on laptop) to get similarity scores
- ▶ Utilizes the Java API for interaction with the NPU and for building of the app.

Demo: Android App Functionalities

- ▶ The ability to upload a picture.
- ▶ Either choose a mask for the image or draw over an area to in-paint depending on the function of the app.
- ▶ Run the model to get:
 - ▶ The image output (for a regular in-painting app)
 - ▶ Benchmarking metrics (for the benchmarking app)
 - ▶ **In Progress**
 - ▶ Built for the app which runs on the CPU
 - ▶ For advanced methods like LPIPS, the benchmarking runs separately on a computer through a python script.

Output of Apps Created/Used

C++ API

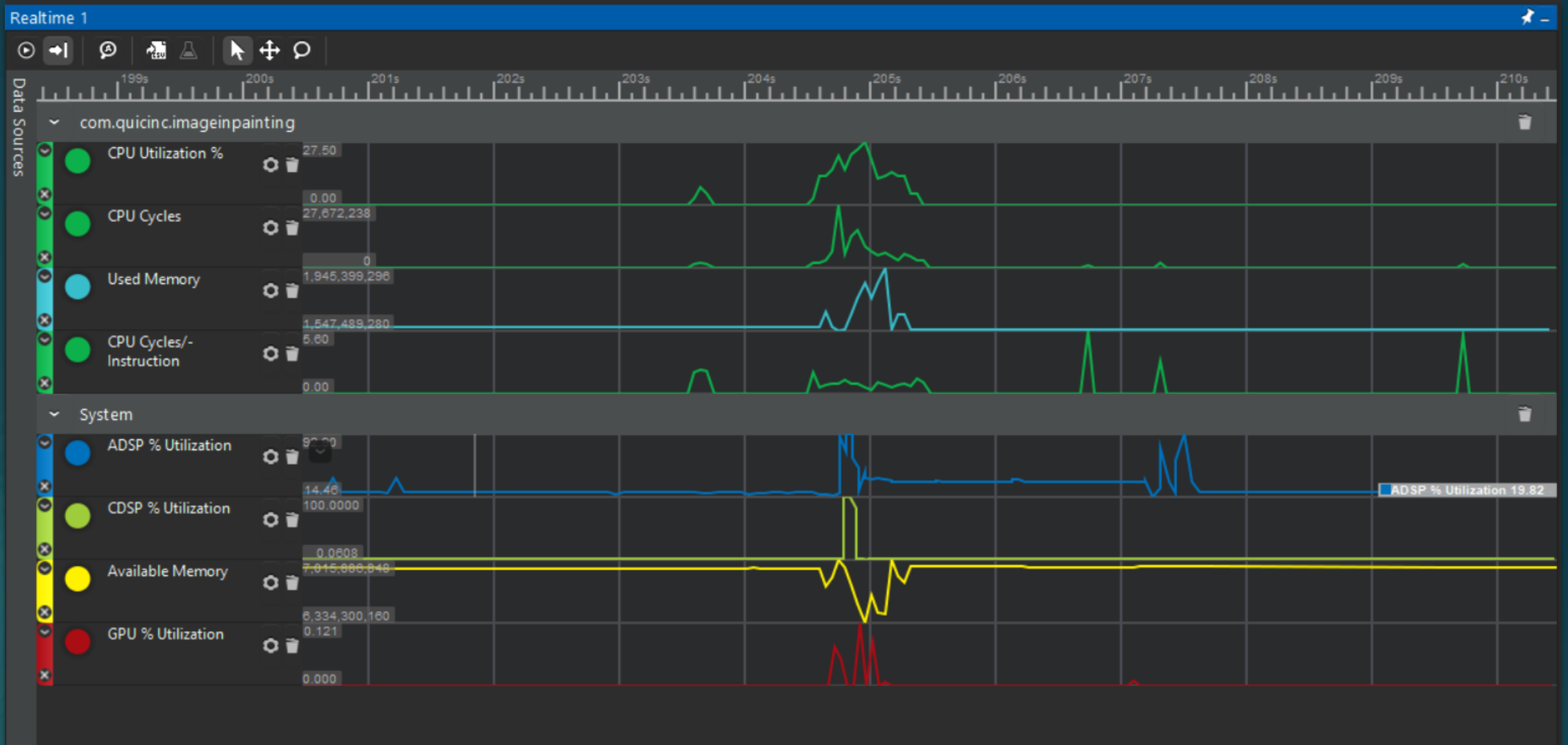
- ▶ Output Image
- ▶ Run time on CPU
- ▶ Memory Usage
- ▶ PSNR
- ▶ SSIM

JAVA API (Pre and Post NPU)

- ▶ Output Image
- ▶ Run time on CPU
- ▶ Run time on GPU
- ▶ Run time on NPU
- ▶ RAM Usage
- ▶ Detailed Benchmarking done using Snapdragon Profiler

SNPE-BENCH.py

- ▶ Output Raws
- ▶ Excel Containing
 - ▶ Respective Runtime
 - ▶ Respective Memory Usage
 - ▶ Load Times and more
- ▶ For Similarity,
 - ▶ Utilize the python file to get:
 - ▶ MSE, PSNR
 - ▶ SSIM
 - ▶ LPIPS



CPU, GPU, NPU Benchmarking

UTILIZED SNAPDRAGON PROFILER TO GET THE USAGE METRICS OF THE CPU, GPU, NPU:

--- Image Quality Evaluation Summary ---

LaMa Dilated SIMILARITY RESULTS WHOLE IMAGE

Ideal Images Folder: /Users/kaushikyadala/Documents/IIIT/Third_Sem/ESW/mass_benchmarking/esw_dataset/ideal
Compared Images Folder: /Users/kaushikyadala/Documents/IIIT/Third_Sem/ESW/mass_benchmarking/lama/inpainted
Processed 101 image pairs.

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AVERAGE METRICS (Higher is better, except for LPIPS & FID)

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Average PSNR: 21.1723

Average SSIM: 0.7621

---- More human like checking

Average LPIPS: 0.1826

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DATASET-LEVEL METRIC (Lower is better)

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FID Score: 135.9633

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DETAILED PER-IMAGE ANALYSIS

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Image Name	PSNR	SSIM	LPIPS
016.jpg	20.6819	0.6356	0.0745
002.jpg	18.0677	0.5282	0.1010
003.jpg	23.6901	0.8985	0.1023
017.png	14.1978	0.3225	0.1138
029.png	17.4057	0.7606	0.3326
015.png	27.9547	0.9596	0.0993
001.png	28.6297	0.9378	0.0577
014.jpg	20.4212	0.6586	0.0989
028.png	20.1921	0.8343	0.2130
010.png	24.2186	0.9304	0.1027
004.jpg	19.9267	0.6376	0.1309
038.jpg	20.6462	0.8641	0.1006
005.jpg	24.1984	0.8360	0.1032
039.png	24.2627	0.9130	0.0903
011.png	24.3576	0.9656	0.1345
007.png	24.1387	0.9511	0.1121
013.jpg	24.3071	0.8378	0.0809
012.png	27.9112	0.9229	0.1128
006.png	19.7587	0.9399	0.2026
101.png	20.7166	0.7196	0.0911
049.png	23.8797	0.9308	0.0959
075.png	17.1183	0.6590	0.3072
061.jpg	14.7109	0.7321	0.3842
074.png	23.3453	0.9276	0.3807
060.png	23.7857	0.9323	0.0649
048.png	25.9307	0.8956	0.2791
100.png	20.5670	0.6649	0.1000
102.png	29.3451	0.9196	0.0670
076.png	9.1458	0.5500	0.4546
089.png	21.4985	0.4867	0.0648
062.jpg	13.6954	0.4205	0.3354
063.jpg	15.5028	0.5426	0.2840
088.png	21.8550	0.5577	0.1462
077.png	12.4660	0.2863	0.4214
073.png	11.8701	0.7692	0.3640
067.png	28.3448	0.7541	0.1160
098.png	27.0780	0.9127	0.0629
099.png	23.4643	0.9290	0.1414
066.png	20.2819	0.4563	0.1974
072.png	17.3881	0.5854	0.2084
070.png	29.0608	0.7393	0.2288
058.png	25.8559	0.9412	0.0358
064.jpg	19.4810	0.6979	0.1824
059.jpg	30.2859	0.9334	0.0823
065.jpg	19.9946	0.6705	0.1836
071.png	20.7320	0.8693	0.2044
068.png	25.8859	0.9238	0.1280
054.png	25.2989	0.9210	0.0609
083.png	18.7739	0.5466	0.2533
097.png	23.8721	0.7910	0.0968
040.jpg	19.4767	0.5099	0.0870
096.png	21.7999	0.8653	0.1110
082.png	14.8816	0.8486	0.5535
055.png	25.8271	0.9614	0.0992
041.png	18.2277	0.8416	0.2428
069.png	18.0494	0.7946	0.2609
057.png	18.5684	0.8501	0.1411
043.png	23.7106	0.9225	0.2094
094.png	27.3201	0.8791	0.0279
080.png	9.2027	0.5163	0.5778
081.png	17.0234	0.8169	0.3206
095.png	20.6479	0.6845	0.1193
042.png	19.4539	0.8306	0.1848
056.png	20.7163	0.9068	0.3105
052.png	25.7370	0.9440	0.1024
046.png	18.3409	0.8361	0.3173
091.png	12.2976	0.1255	0.2218
085.png	17.0021	0.7186	0.4126
084.png	28.4168	0.7585	0.2095
090.png	14.5472	0.3455	0.1005
047.png	28.7580	0.9761	0.0848
053.png	28.4499	0.9568	0.0432
045.png	13.8732	0.7166	0.4180
051.png	26.2180	0.9423	0.0376
079.png	17.0244	0.6606	0.4679
086.png	19.9698	0.8550	0.2864
092.png	21.1602	0.6401	0.1570
093.png	22.6465	0.6369	0.1427

087.png	22.1569	0.8788	0.0547
078.png	15.9278	0.7721	0.2960
050.png	30.6815	0.9606	0.0310
044.png	25.7201	0.8856	0.2081
023.png	24.9746	0.9372	0.0889
037.png	25.4981	0.9473	0.1328
022.jpg	14.1725	0.4542	0.1145
036.jpg	18.4594	0.2276	0.2026
008.png	24.8204	0.9087	0.1284
034.png	20.2800	0.8586	0.1484
020.jpg	22.0129	0.4833	0.0899
021.jpg	21.0017	0.4454	0.1254
035.png	20.2038	0.9090	0.2810
009.png	23.1120	0.9441	0.0840
019.png	27.0761	0.9564	0.2221
031.jpg	26.9332	0.9346	0.0463
030.jpg	16.2767	0.4202	0.1251
024.jpg	20.4817	0.6438	0.0883
018.png	16.3299	0.8914	0.4182
032.png	19.5851	0.9173	0.1663
026.jpg	16.5165	0.6100	0.1459
033.png	13.6285	0.7849	0.2069
027.png	17.0050	0.8571	0.3990

--- Image Quality Evaluation Summary ---

LaMa Dilated SIMILARITY RESULTS WITHIN MASK

Ideal Images Folder: /Users/kaushikyadala/Documents/IIIT/Third_Sem/ESW/mass_benchmarking/esw_dataset/ideal
Compared Images Folder: /Users/kaushikyadala/Documents/IIIT/Third_Sem/ESW/mass_benchmarking/lama/inpainted
Processed 101 image pairs.

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AVERAGE METRICS (Higher is better, except for LPIPS & FID)

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Average PSNR: 14.7645
Average SSIM: 0.7636
Average LPIPS: 0.0733

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DATASET-LEVEL METRIC (Lower is better)

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FID Score: 135.9633

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DETAILED PER-IMAGE ANALYSIS

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Image Name	PSNR	SSIM	LPIPS
016.jpg	20.4814	0.6177	0.0373
002.jpg	19.0780	0.5183	0.0084
003.jpg	11.9248	0.8959	0.0104
017.png	12.5383	0.3232	0.2065
029.png	13.8209	0.7681	0.2188
015.png	16.8369	0.9636	0.0123
001.png	19.0608	0.9608	0.0357
014.jpg	18.5814	0.6489	0.0171
028.png	13.8781	0.8481	0.1219
010.png	15.4223	0.9434	0.0379
004.jpg	13.6529	0.6218	0.0689
038.jpg	12.1013	0.8822	0.0949
005.jpg	16.5877	0.8359	0.0756
039.png	12.7604	0.9324	0.0361
011.png	15.0544	0.9688	0.0230
007.png	15.5602	0.9588	0.0358
013.jpg	17.3296	0.8434	0.0452
012.png	19.7562	0.9322	0.0431
006.png	12.1484	0.9409	0.0492
101.png	16.8748	0.7178	0.0506
049.png	15.7583	0.9394	0.0479
075.png	16.0466	0.6607	0.1608
061.jpg	11.9797	0.7317	0.0778
074.png	17.8913	0.9272	0.0493
060.png	14.2926	0.9407	0.0307
048.png	21.4229	0.8956	0.1186
100.png	13.4572	0.6591	0.0519
102.png	19.2954	0.9219	0.0130
076.png	5.8885	0.5482	0.2330
089.png	18.7825	0.4867	0.0022
062.jpg	10.5187	0.4197	0.1203
063.jpg	9.9901	0.5399	0.0909
088.png	16.2125	0.5577	0.0280
077.png	10.4597	0.2831	0.2167
073.png	6.3219	0.7689	0.1195
067.png	22.8601	0.7495	0.0088
098.png	18.1252	0.9294	0.0396
099.png	15.0096	0.9321	0.0331
066.png	9.7010	0.4466	0.0254
072.png	10.0449	0.5774	0.0242
070.png	21.6454	0.7380	0.0191
058.png	13.2131	0.9589	0.0367
064.jpg	7.7752	0.6924	0.0311
059.jpg	21.1451	0.9414	0.0180
065.jpg	10.0402	0.6655	0.0179
071.png	10.5903	0.8687	0.0522
068.png	26.2607	0.9232	0.0061
054.png	15.3395	0.9372	0.0347
083.png	12.1185	0.5438	0.0224
097.png	16.9875	0.7910	0.0516
040.jpg	11.0838	0.4752	0.0194
096.png	14.1927	0.8653	0.0357
082.png	11.1565	0.8486	0.1767
055.png	14.2952	0.9667	0.0173
041.png	13.3031	0.8504	0.1071
069.png	11.5119	0.8014	0.0967
057.png	11.2046	0.8658	0.1059
043.png	18.9234	0.9242	0.0968
094.png	16.5410	0.8791	0.0049
080.png	7.4026	0.5137	0.3705
081.png	11.4952	0.8158	0.0886
095.png	13.4173	0.6845	0.0453
042.png	12.7644	0.8423	0.1669
056.png	15.3875	0.9086	0.0383
052.png	16.7587	0.9490	0.0602
046.png	13.7846	0.8385	0.1139
091.png	9.9772	0.1255	0.1652
085.png	14.5370	0.7184	0.0631
084.png	17.5158	0.7502	0.0165
090.png	8.7907	0.3455	0.0749
047.png	16.9287	0.9788	0.0123
053.png	16.1351	0.9696	0.0220
045.png	10.6178	0.7157	0.2215
051.png	13.5011	0.9542	0.0085
079.png	17.6325	0.6582	0.1990
086.png	14.6856	0.8637	0.0931
092.png	15.7377	0.6401	0.1358
093.png	17.7238	0.6369	0.1269
087.png	11.6180	0.8788	0.0342
078.png	12.8310	0.7720	0.1122

050.png	17.1216	0.9737	0.0169
044.png	21.3709	0.8908	0.0993
023.png	14.5272	0.9427	0.0184
037.png	16.6788	0.9515	0.0290
022.jpg	15.4150	0.4456	0.0074
036.jpg	14.6670	0.2100	0.1006
008.png	17.7039	0.9152	0.0638
034.png	12.7118	0.8633	0.0958
020.jpg	19.4039	0.4669	0.0187
021.jpg	16.4402	0.4282	0.0509
035.png	15.5217	0.9135	0.1280
009.png	13.3253	0.9542	0.0401
019.png	21.8826	0.9590	0.0508
031.jpg	16.7438	0.9421	0.0570
030.jpg	15.2719	0.4138	0.0706
024.jpg	20.4703	0.6334	0.0190
018.png	12.2109	0.8945	0.1126
032.png	12.5136	0.9241	0.1124
026.jpg	12.3188	0.6035	0.0696
033.png	7.9786	0.7988	0.1788
027.png	12.8590	0.8610	0.2219

--- Image Quality Evaluation Summary ---

AOT_GAN SIMILARITY RESULTS WHOLE IMAGE

Ideal Images Folder: /Users/kaushikyadala/Documents/IIIT/Third_Sem/ESW/mass_benchmarking/esw_dataset/ideal
Compared Images Folder: /Users/kaushikyadala/Documents/IIIT/Third_Sem/ESW/mass_benchmarking/aot_gan/inpainted
Processed 101 image pairs.

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AVERAGE METRICS (Higher is better, except for LPIPS & FID)

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Average PSNR: 21.1251

Average SSIM: 0.7579

---- More human like checking

Average LPIPS: 0.1903

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DATASET-LEVEL METRIC (Lower is better)

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FID Score: 138.4853

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DETAILED PER-IMAGE ANALYSIS

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Image Name	PSNR	SSIM	LPIPS
016.jpg	19.7407	0.6224	0.0997
002.jpg	17.9349	0.5286	0.1001
003.jpg	31.3862	0.9351	0.0161
017.png	12.2926	0.2598	0.1671
029.png	13.1321	0.7499	0.4145
015.png	30.1733	0.9663	0.0731
001.png	25.0243	0.9234	0.0516
014.jpg	20.4025	0.6630	0.1031
020.png	20.8483	0.8305	0.1836
010.png	26.1377	0.9340	0.1121
004.jpg	20.7474	0.6389	0.1281
038.jpg	21.7376	0.8640	0.1043
005.jpg	22.1632	0.8310	0.1256
039.png	24.5047	0.9172	0.0853
011.png	23.6367	0.9673	0.1272
007.png	21.1954	0.9491	0.1182
013.jpg	21.0862	0.8260	0.1005
012.png	28.3104	0.9214	0.1105
006.png	26.3553	0.9498	0.1654
101.png	19.7055	0.7162	0.1179
049.png	21.2270	0.9301	0.1228
075.png	14.6885	0.6298	0.3202
061.jpg	14.9066	0.7194	0.3849
074.png	23.6367	0.9217	0.3913
060.png	24.9795	0.9396	0.0590
048.png	15.4222	0.8024	0.3806
100.png	20.4120	0.6715	0.1175
102.png	23.7617	0.9122	0.0758
076.png	11.0807	0.5785	0.4442
089.png	21.5449	0.4892	0.0473
062.jpg	13.8668	0.4125	0.3410
063.jpg	18.0933	0.5576	0.2973
088.png	20.7821	0.5635	0.1346
077.png	14.1708	0.2816	0.4247
073.png	13.9937	0.7784	0.3657
067.png	27.2929	0.7335	0.1380
098.png	23.3896	0.9100	0.0693
099.png	23.8707	0.9250	0.1466
066.png	19.7799	0.4242	0.2979
072.png	17.3705	0.5663	0.2651
070.png	28.2314	0.7263	0.2345
058.png	25.8995	0.9399	0.0379
064.jpg	21.1230	0.7025	0.1648
059.jpg	27.8252	0.9360	0.0597
065.jpg	19.5717	0.6163	0.5069
071.png	22.7411	0.8720	0.1985
068.png	25.6160	0.9175	0.1411
054.png	24.5189	0.9212	0.0736
083.png	18.9341	0.5418	0.2786
097.png	19.2093	0.7669	0.1284
040.jpg	20.2809	0.5116	0.0824
096.png	22.3679	0.8842	0.0820
082.png	17.1031	0.8373	0.4859
055.png	30.8908	0.9673	0.0792
041.png	19.6548	0.8425	0.2858
069.png	23.3088	0.8199	0.2572
057.png	14.6304	0.8534	0.1437
043.png	17.4309	0.8833	0.2916
094.png	27.8085	0.8777	0.0164
080.png	11.7736	0.5699	0.4946
081.png	20.7372	0.8138	0.2729
095.png	21.6702	0.6980	0.1107
042.png	18.8992	0.8355	0.1771
056.png	25.7418	0.9201	0.3154
052.png	24.2457	0.9373	0.0969
046.png	18.2818	0.8349	0.3197
091.png	12.8280	0.1219	0.2299
085.png	18.1597	0.7216	0.3854
084.png	28.5971	0.7304	0.2404
090.png	15.6140	0.3517	0.0862
047.png	28.1409	0.9731	0.0968
053.png	25.3953	0.9476	0.0564
045.png	17.7996	0.7325	0.4112
051.png	28.9627	0.9421	0.0329
079.png	16.0401	0.6191	0.4204
086.png	20.5326	0.8484	0.2946
092.png	22.1057	0.6491	0.1803
093.png	20.1983	0.6396	0.1874

087.png	24.3929	0.8815	0.0509
078.png	16.0673	0.7516	0.3027
050.png	29.1846	0.9592	0.0327
044.png	16.4189	0.8499	0.2541
023.png	29.5206	0.9431	0.0634
037.png	25.6147	0.9462	0.1469
022.jpg	14.1971	0.4591	0.1099
036.jpg	18.3802	0.2352	0.1898
008.png	21.2301	0.9002	0.1220
034.png	19.4749	0.8479	0.1595
020.jpg	22.2568	0.4843	0.1056
021.jpg	21.1392	0.4416	0.1342
035.png	21.8710	0.9066	0.2803
009.png	26.9714	0.9545	0.0542
019.png	18.0185	0.8986	0.3548
031.jpg	25.4890	0.9395	0.0424
030.jpg	15.6582	0.4168	0.1319
024.jpg	19.4850	0.6361	0.1299
018.png	20.2098	0.8885	0.3554
032.png	21.4070	0.9190	0.1795
026.jpg	16.6904	0.6098	0.1372
033.png	18.5988	0.7734	0.2227
027.png	17.7015	0.8603	0.3755

--- Image Quality Evaluation Summary ---

AOT GAN SIMILARITY RESULTS WITHIN MASK

Ideal Images Folder: /Users/kaushikyadala/Documents/IIIT/Third_Sem/ESW/mass_benchmarking/esw_dataset/ideal
Compared Images Folder: /Users/kaushikyadala/Documents/IIIT/Third_Sem/ESW/mass_benchmarking/aot_gan/inpainted
Processed 101 image pairs.

=====

AVERAGE METRICS (Higher is better, except for LPIPS & FID)

=====

Average PSNR: 15.1341
Average SSIM: 0.7591
Average LPIPS: 0.0750

=====

DATASET-LEVEL METRIC (Lower is better)

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FID Score: 138.4853

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DETAILED PER-IMAGE ANALYSIS

=====

Image Name	PSNR	SSIM	LPIPS
016.jpg	14.0465	0.6042	0.0511
002.jpg	17.9211	0.5182	0.0073
003.jpg	32.6013	0.9321	0.0001
017.png	8.0241	0.2601	0.2476
029.png	9.5382	0.7570	0.3054
015.png	19.2626	0.9705	0.0025
001.png	15.2870	0.9472	0.0350
014.jpg	19.0343	0.6528	0.0186
028.png	14.5784	0.8438	0.1007
010.png	17.5622	0.9466	0.0403
004.jpg	15.6965	0.6222	0.0636
038.jpg	13.3959	0.8803	0.0771
005.jpg	13.8109	0.8300	0.0955
039.png	13.0357	0.9362	0.0373
011.png	14.3186	0.9705	0.0262
007.png	12.5869	0.9567	0.0526
013.jpg	13.1386	0.8304	0.0626
012.png	20.1854	0.9307	0.0391
006.png	19.4715	0.9506	0.0292
101.png	13.1696	0.7144	0.0715
049.png	13.0344	0.9387	0.0666
075.png	10.0672	0.6313	0.1649
061.jpg	12.7809	0.7191	0.0829
074.png	18.6841	0.9216	0.0511
060.png	15.6096	0.9482	0.0300
048.png	10.5596	0.8021	0.1722
100.png	13.1459	0.6657	0.0597
102.png	12.0198	0.9142	0.0162
076.png	10.3922	0.5762	0.2259
089.png	22.7053	0.4892	0.0027
062.jpg	11.3183	0.4119	0.1023
063.jpg	17.3727	0.5544	0.0789
088.png	13.9305	0.5635	0.0417
077.png	14.2122	0.2780	0.2178
073.png	11.1270	0.7781	0.0972
067.png	16.0366	0.7281	0.0080
098.png	13.8139	0.9261	0.0548
099.png	15.4494	0.9279	0.0414
066.png	13.7465	0.4125	0.0141
072.png	12.3145	0.5570	0.0190
070.png	17.4917	0.7244	0.0129
058.png	13.3515	0.9570	0.0438
064.jpg	15.6615	0.6967	0.0124
059.jpg	17.8710	0.9436	0.0238
065.jpg	12.1885	0.6085	0.0106
071.png	19.0573	0.8715	0.0266
068.png	16.5161	0.9166	0.0096
054.png	14.4025	0.9369	0.0423
083.png	17.7070	0.5387	0.0145
097.png	10.2943	0.7666	0.0749
040.jpg	14.4929	0.4767	0.0158
096.png	16.0531	0.8842	0.0277
082.png	14.4252	0.8371	0.1304
055.png	22.0646	0.9724	0.0069
041.png	14.7658	0.8508	0.1452
069.png	18.0028	0.8259	0.0661
057.png	7.1166	0.8689	0.1121
043.png	12.6138	0.8855	0.1424
094.png	19.8370	0.8777	0.0028
080.png	11.2603	0.5670	0.2705
081.png	17.2690	0.8125	0.0767
095.png	15.0610	0.6980	0.0552
042.png	12.2147	0.8466	0.1673
056.png	20.5819	0.9221	0.0368
052.png	15.1954	0.9423	0.0593
046.png	13.7552	0.8375	0.1116
091.png	11.3942	0.1219	0.1679
085.png	19.0478	0.7215	0.0662
084.png	23.5717	0.7194	0.0060
090.png	12.3529	0.3517	0.0607
047.png	16.3130	0.9757	0.0128
053.png	12.6153	0.9598	0.0282
045.png	14.9571	0.7315	0.1870
051.png	17.7056	0.9534	0.0042
079.png	14.4525	0.6166	0.1693
086.png	15.5111	0.8572	0.1028
092.png	16.9601	0.6491	0.1367
093.png	14.5904	0.6396	0.1602
087.png	14.8177	0.8815	0.0292
078.png	13.2694	0.7512	0.1244

050.png	15.3357	0.9720	0.0177
044.png	10.2418	0.8554	0.1193
023.png	19.8118	0.9483	0.0081
037.png	16.7777	0.9501	0.0374
022.jpg	16.1512	0.4511	0.0043
036.jpg	14.6310	0.2178	0.1070
008.png	13.9962	0.9066	0.0619
034.png	11.7295	0.8525	0.0975
020.jpg	21.0176	0.4671	0.0255
021.jpg	16.9393	0.4234	0.0637
035.png	17.2437	0.9107	0.1421
009.png	17.7153	0.9644	0.0335
019.png	12.7774	0.9009	0.1573
031.jpg	15.1001	0.9461	0.0308
030.jpg	11.7752	0.4099	0.0884
024.jpg	15.0201	0.6254	0.0321
018.png	16.1445	0.8913	0.1163
032.png	14.5508	0.9255	0.1073
026.jpg	12.9867	0.6029	0.0669
033.png	13.2252	0.7842	0.1803
027.png	13.5752	0.8642	0.2152

=====

IMAGE CLASSIFICATION GROUPS

=====

COMPLEXITY (Based on Edge Density & Texture)

High Complexity (57 images):

- 007
- 008
- 009
- 010
- 012
- 015
- 017
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- 023
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- 098
- 099
- 100
- 101

Medium Complexity (21 images):

- 006
- 011
- 019
- 037
- 043
- 048
- 056
- 064
- 067
- 070
- 071
- 072
- 073
- 075
- 087
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- 092
- 093
- 095
- 096
- 102

Low Complexity (24 images):

- 001
- 002
- 003
- 004
- 005
- 013
- 014
- 016
- 020
- 021
- 024
- 025
- 026
- 030
- 031

- 036
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- 040
- 059
- 068
- 069
- 084
- 089
- 094

BRIGHTNESS

Very Bright (2 images):

- 022
- 082

Bright (37 images):

- 002
- 003
- 005
- 006
- 007
- 009
- 011
- 012
- 014
- 015
- 018
- 021
- 023
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- 098
- 099
- 100

Medium (56 images):

- 004
- 008
- 010
- 013
- 016
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- 032
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- 084
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- 090
- 091
- 097
- 101
- 102

Dark (7 images):

- 001
- 017
- 067
- 068
- 071
- 092
- 093

=====

SATURATION

High Saturation (26 images):

- 001
- 002
- 010
- 012
- 017
- 019
- 020
- 046
- 062
- 065
- 066
- 067
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- 095
- 096

Medium Saturation (52 images):

- 003
- 006
- 007
- 008
- 011
- 018
- 021
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- 028
- 029
- 030
- 033
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- 102

Low Saturation (24 images):

- 004
- 005
- 009
- 013
- 014
- 015
- 016
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- 026
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- 089
- 097
- 101

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MASK COVERAGE

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Medium Coverage (25 images):

- 018
- 019
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- 093

Small Coverage (77 images):

- 001
- 002
- 003
- 004
- 005
- 006
- 007
- 008
- 009
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- 011
- 012
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MASK LOCATION

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Center (51 images):

- 001
- 005
- 006
- 017
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Edge/Corner (51 images):

- 002
- 003
- 004
- 007
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MASK FRAGMENTATION

Single Region (97 images):

- 001
- 002
- 003
- 004
- 005
- 006
- 007
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Few Regions (5 images):

- 030
- 052
- 056
- 077
- 086

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CONTENT	TYPE
Contains Faces (13 images):	
- 001	
- 017	
- 020	
- 024	
- 026	
- 034	
- 035	
- 043	
- 049	
- 087	
- 088	
- 090	
- 091	
Contains Text (7 images):	
- 016	
- 024	
- 050	
- 061	
- 073	
- 075	
- 092	
General (83 images):	
- 002	
- 003	
- 004	
- 005	
- 006	
- 007	
- 008	
- 009	
- 010	
- 011	
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Contains Text (7 images):

General (83 images):

- 038
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Additional Achievements

- ▶ Got another inpainting model called AOT-GAN to run on the NPU.
- ▶ Created a 101 image dataset, with images, masks, and ideal images.
- ▶ Classified all our images based on the following criteria:
 - Resolution & aspect ratio - Basic image dimensions
 - Face detection - Uses MediaPipe to count faces
 - Texture analysis - Laplacian variance (blur detection) and edge density
 - Color analysis - Brightness, saturation, and dominant colors via k-means clustering
 - Text detection - Uses Tesseract OCR
 - CLIP embeddings - Semantic content understanding using OpenAI's CLIP model
 - Mask analysis - If a mask image is provided, it analyzes mask area, bounding box, location, and fragmentation
- ▶ Working on trying to utilize the above classifications to create a model selector (suggested by qualcomm)
- ▶ Working on getting another model that is not on the Qualcomm AI Hub to run on the NPU after optimizing and quantizing it for the NPU and converting the model into a TF-Lite.

Model Creation

- ▶ Created a simple model based on the similarity score results only, fine tuned image res net 18.
 - The choosing scheme is:
 - LPIPS > SSIM > PSNR
- ▶ Model Matrix:

--- Final Evaluation ---				
	precision	recall	f1-score	support
Model A Wins	0.88	0.64	0.74	11
Model B Wins	0.69	0.90	0.78	10
accuracy			0.76	21
macro avg	0.78	0.77	0.76	21
weighted avg	0.79	0.76	0.76	21

Future Plans

- ▶ We have completed all the deliverables mentioned by Qualcomm for our project.
- ▶ Further suggestions on how to proceed to extra stuff by Qualcomm:
 - Create a pipeline such that the app chooses one of the models we have in our app based on what is being in-painted in the image. We need to choose the model that will perform best for the given image.
 - Try running the AI models for measuring the similarity of the inpainted images on the app itself.

Image 1 (LaMa-Dilated)

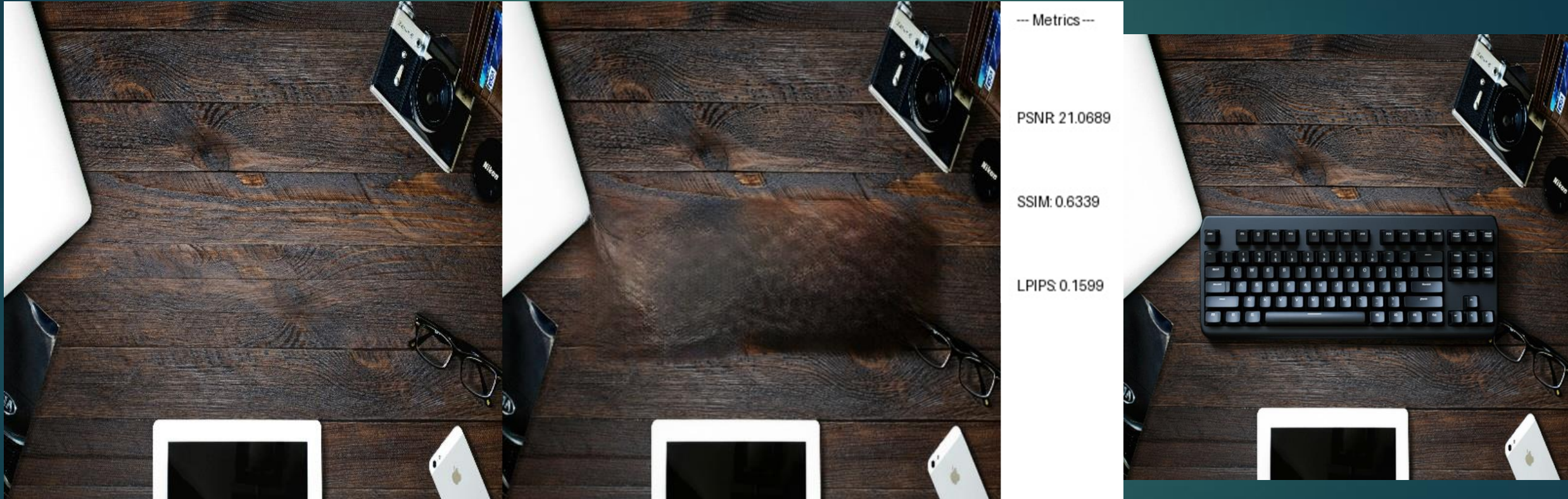


Image 2 (LaMa-Dilated)



--- Metrics ---

PSNR 20.5913

SSIM: 0.6833

LPIPS 0.1190



Image 3 (LaMa-Dilated)



--- Metrics ---

PSNR 22.1182

SSIM: 0.8796

LPIPS: 0.0555



Image 4 (LaMa-Dilated)



--- Metrics ---

PSNR 21.5627

SSIM: 0.4862

LPIPS 0.0648

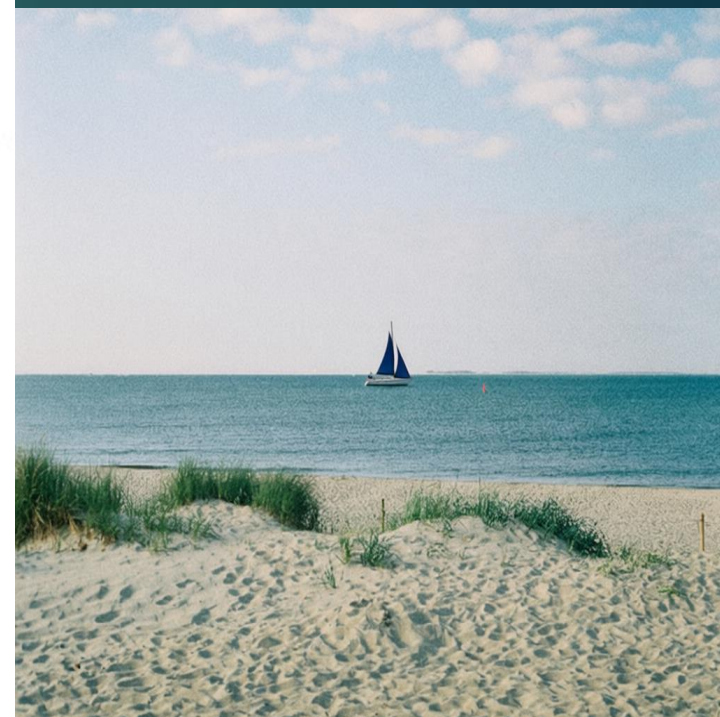


Image 5 (LaMa-Dilated)



--- Metrics ---

PSNR 21.8867

SSIM: 0.5569

LPIPS: 0.1446



Image 6 (LaMa-Dilated)



--- Metrics ---

PSNR: 27.2867

SSIM: 0.8762

LPIPS: 0.0280

